DO NOW — RECALL

Five process scheduling algorithms

Name: _____ Class: _____ Date: _____

1. What is a Ready process?

2. What is a scheduling quantum?

3. What is a context switch?

4. Sorted [1,3,5,7,9,11,13,15,17], target 15. Trace midpoint indexes.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Eligible for CPU time.

2. Model answer

Maximum CPU time per RR turn.

3. Model answer

Saving and restoring execution context when changing process.

4. Model answer

mid4→9, mid6→13, mid7→15 found; indexes 4,6,7.

Vocabulary — know these terms

FCFS — First Come First Served.

SJF — Shortest Job First.

SRT — Shortest Remaining Time.

RR — Round Robin.

MLFQ — Multilevel feedback queue.

NEW LEARNING

Read, annotate and explain

Understand

FCFS serves arrival order; SJF selects the shortest available complete job.

Trace

SRT can preempt when a newly ready job has shorter remaining time; RR cycles by quantum.

Evaluate

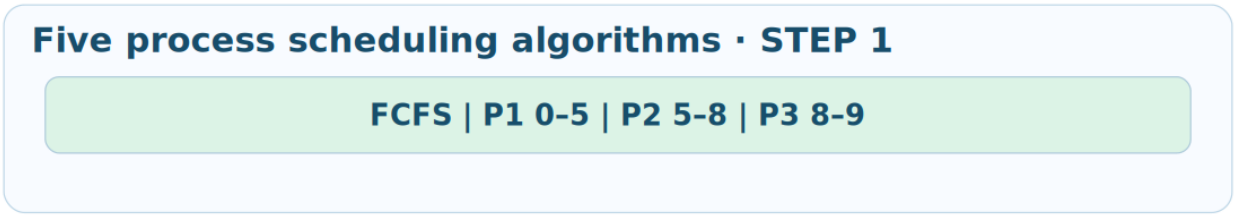
Multilevel feedback queues adapt priority by task behaviour. Compare responsiveness, waits, starvation and context-switch overhead.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

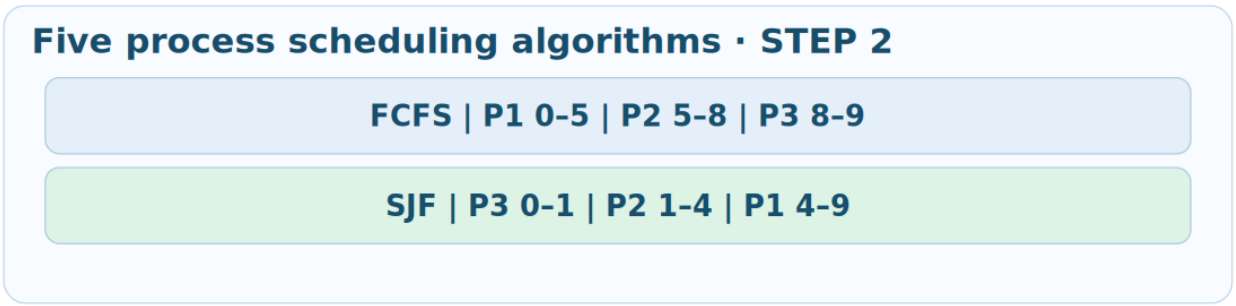
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



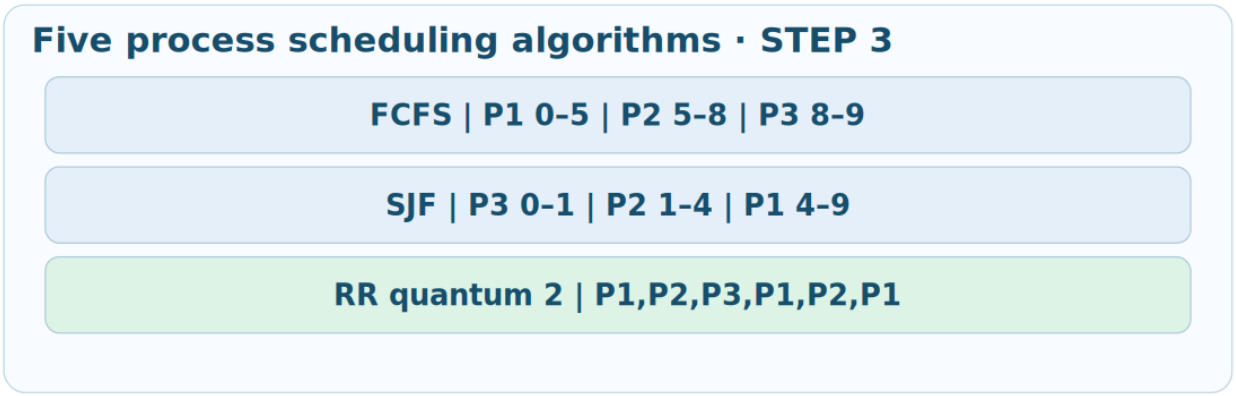
Five process scheduling algorithms · STEP 1

New Learning



Five process scheduling algorithms · STEP 2

New Learning

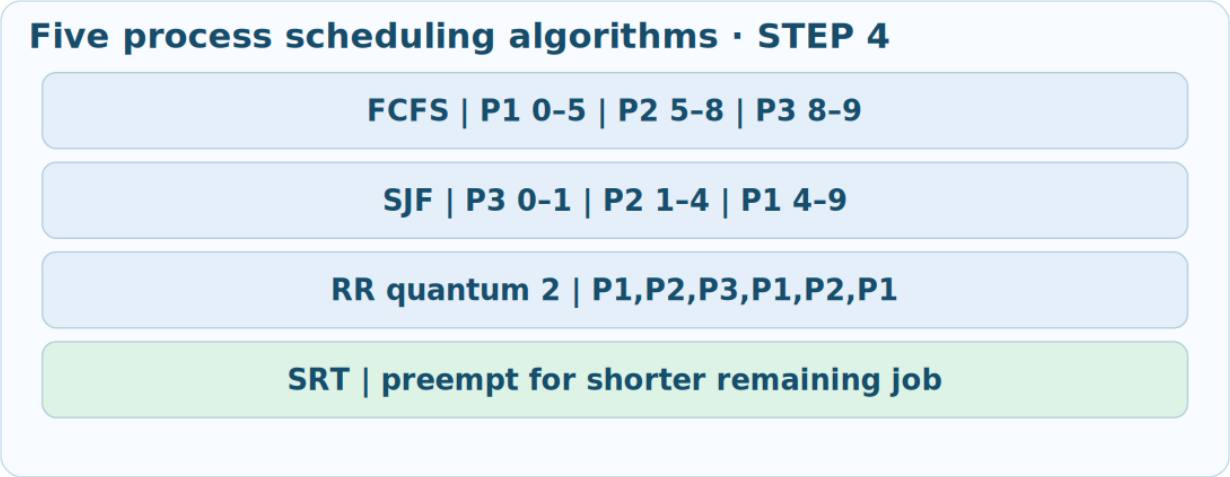


Five process scheduling algorithms · STEP 3

New Learning · original lesson diagrams

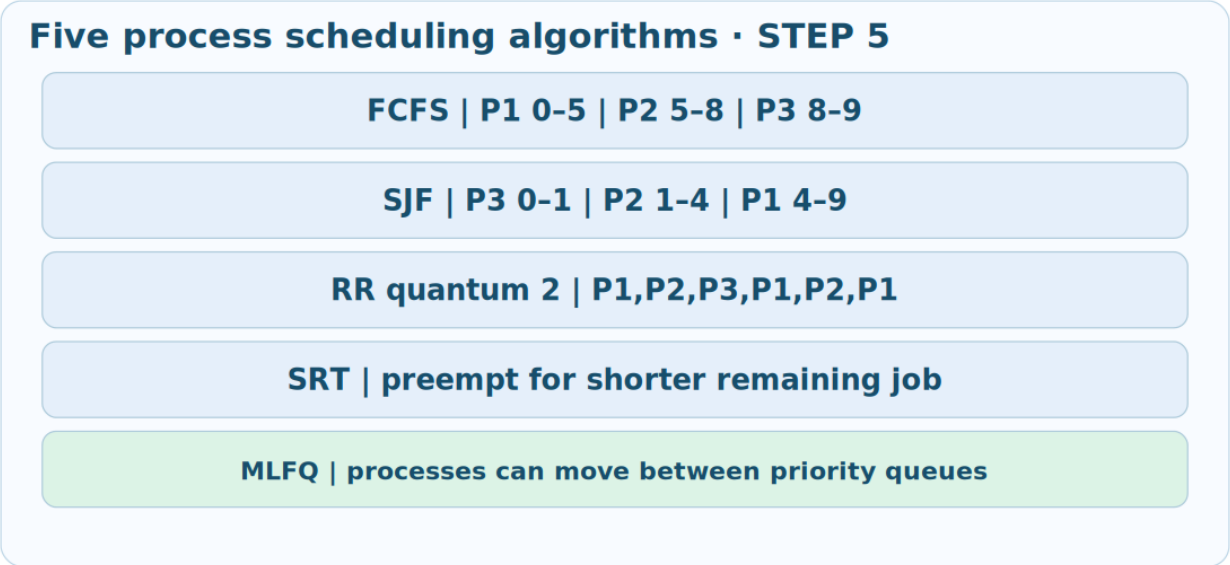
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Five process scheduling algorithms · STEP 4

New Learning



Five process scheduling algorithms · STEP 5

I DO — WORKED EXAMPLE

Follow the worked method

A=4,B=2 arrive together: trace non-preemptive SJF.

Predict first

Shorter job B runs first

Model and check

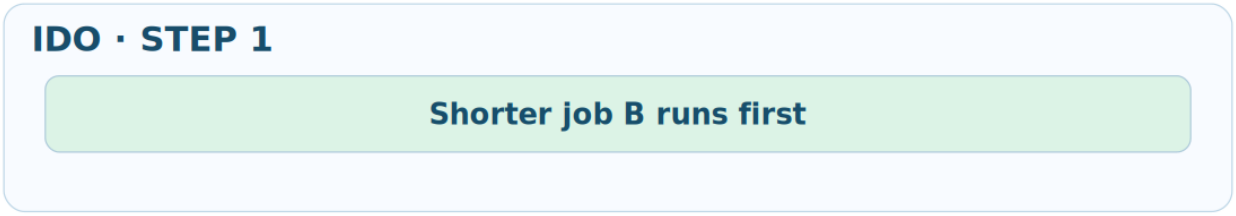
A waits two time units.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A=4,B=2 arrive together: trace non-preemptive SJF.

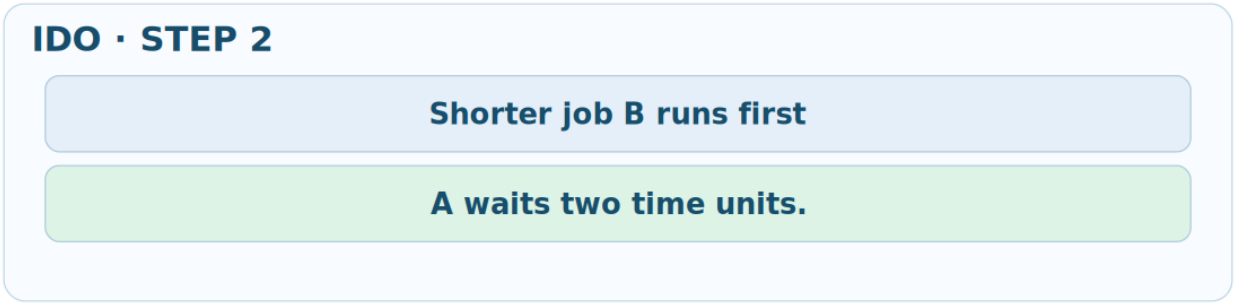
Shorter job B runs first



IDO · STEP 1

A=4,B=2 arrive together: trace non-preemptive SJF.

Shorter job B runs first



IDO · STEP 2

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A=3,B=2,C=1 all at time zero: trace FCFS and waits. Break simultaneous-arrival ties in the listed A, B, C orde

Arrival order A,B,C gives mean waiting (0+3+5)/3=8/3.



WEDO · STEP 1

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Arrival order A,B,C gives mean waiting (0+3+5)/3=8/3.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Name FCFS,SJF,SRT,RR and multilevel feedback by defining feature. [5]

Question 2

A=5,B=3,C=1 all arrive at t=0. For FCFS, break ties in A, B, C order; for non-preemptive SJF, select the shortest available job. Compare the two orders and mean waiting times. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Using Round Robin with time quantum 2, processes A=4, B=3 and C=1 all arrive at t=0; take the initial queue order as A, B, C. Give all execution slices and one drawback of a short quantum. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Round Robin always lets one process finish before any other runs.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

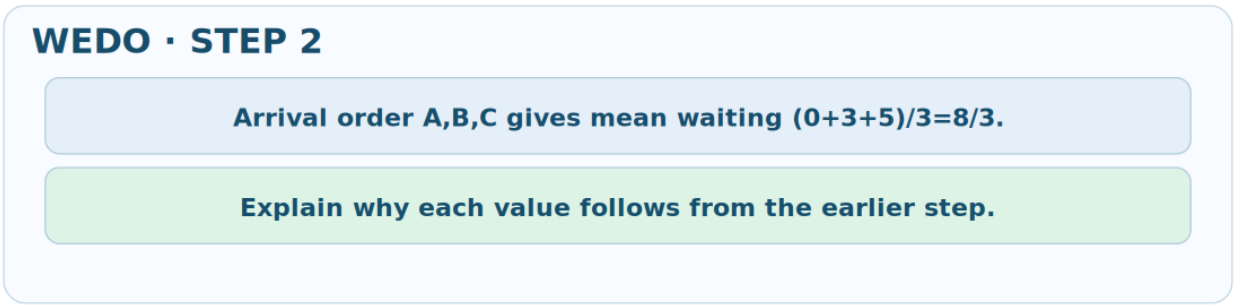
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A=3,B=2,C=1 all at time zero: trace FCFS and waits. Break simultaneous-arrival ties in the listed A, B, C orde

Arrival order A,B,C gives mean waiting (0+3+5)/3=8/3.



WEDO · STEP 2

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Explain why each value follows from the earlier step.

IMPROVE: Explain one correction to your own reasoning.